

Financial Vault

Executable Deliverable — Installation & Run Guide

Project 11 · Information Security · Faculty of Engineering

1. What Is Delivered

The executable deliverable is a fully functional Python/Flask web application. It runs locally on any machine with Python 3.10 or newer installed. No internet connection, cloud service, or database server is required. All cryptographic operations happen locally.

File / Folder	Purpose
app.py	Main Flask application — all HTTP routes
crypto.py	Encryption, decryption, hashing logic
audit.py	Audit log database (SQLite)
lockout.py	Failed-attempt tracking and file lockout
templates/base.html	Shared page layout and navigation
templates/index.html	Main vault page (encrypt + decrypt forms)
templates/audit.html	Audit log page with filters
templates/result.html	Result/error page after operations
static/style.css	All visual styling (Berry & Cream theme)
vault/	Auto-created folder storing .enc files and audit.db

2. Prerequisites

Requirement	Minimum Version	How to Check
Python	3.10	<code>python --version</code>
pip	22.0	<code>pip --version</code>
Flask	2.3	<code>pip show flask</code>
PyCryptodome	3.18	<code>pip show pycryptodome</code>

3. Installation

Step 1 — Extract the project

Extract the delivered zip file. You should see the structure listed in Section 1.

Step 2 — Open a terminal in the project folder

On Windows: open PowerShell or Command Prompt, then navigate to the project folder:

```
cd C:\path\to\finalproject
```

Step 3 — Install dependencies

Run the following command to install Flask and PyCryptodome:

```
pip install flask pycryptodome
```

If you are using a virtual environment (recommended), activate it first, then run the command above.

Step 4 — Run the application

```
python app.py
```

Flask will start a development server. You will see output similar to:

```
* Running on http://127.0.0.1:5000 * Debug mode: on
```

Step 5 — Open the browser

Open any browser and go to: `http://127.0.0.1:5000`

The Financial Vault interface will load. No login is required.

4. How to Use the Application

Encrypting a file

- On the main Vault page, click 'Choose File' under the Encrypt section and select any document.
- Select an algorithm from the dropdown (AES-256-CBC is recommended).
- Enter a password and click 'Encrypt & Vault'.
- The result page shows the SHA-256 fingerprint of the original file. The encrypted file appears in the Vault Contents table.

Decrypting a file

- On the main Vault page, select the file from the dropdown under the Decrypt section.
- Enter the correct password and click 'Decrypt & Download'.
- If the password is correct and the file has not been tampered with, the original file is downloaded.

- After 5 wrong passwords, the file is locked. Unlock it from the Audit Log page.

Verifying integrity without downloading

Use the Verify form (on the full version of the project) to confirm the SHA-256 hash of a vaulted file without downloading its contents. This is useful for compliance checks.

Viewing the Audit Log

Click 'Audit Log' in the navigation bar. Every operation is listed with its timestamp, action, file, algorithm, status (success / failed / tampered / blocked), and details. Use the filter buttons to narrow by action type or status.

5. Feature Summary

Feature	Description
Multi-algorithm encryption	AES-256-CBC (recommended), 3DES (legacy), ChaCha20 (modern stream cipher)
PBKDF2 key derivation	100,000 iterations of HMAC-SHA256 — password never used as key directly
Random salt + IV	Fresh randomness per encryption — same file + same password = different ciphertext
SHA-256 integrity	Hash stored in bundle — any tampering detected on decryption
Lockout protection	File locked after 5 wrong passwords — manual unlock from audit log
Persistent audit log	SQLite database records every action with timestamp and status
Self-describing bundles	Algorithm, salt, IV all stored inside the .enc file — no external index
Zero external dependencies	No cloud, no accounts, no external database — fully self-contained

The application is delivered as a ready-to-run Python project. No compilation step is required. All cryptographic operations are performed locally using PyCryptodome, a well-audited open-source library. The vault directory and audit database are created automatically on first run.